

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
```

```
In [2]: df = pd.read_csv('house.csv')
df.head()
```

```
Out[2]:
```

	House Size	House Price
0	1500	350000
1	1800	400000
2	2000	450000
3	2200	475000
4	2500	525000

```
In [3]: X = df['House Size'].to_frame()
Y = df['House Price']
```

```
In [4]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size
```

```
In [5]: model = LinearRegression()
model.fit(X_train, Y_train)
```

```
Out[5]:
```

▼ LinearRegression ⓘ ?

► Parameters

```
In [6]: Y_pred = model.predict(X_test)
Y_pred
```

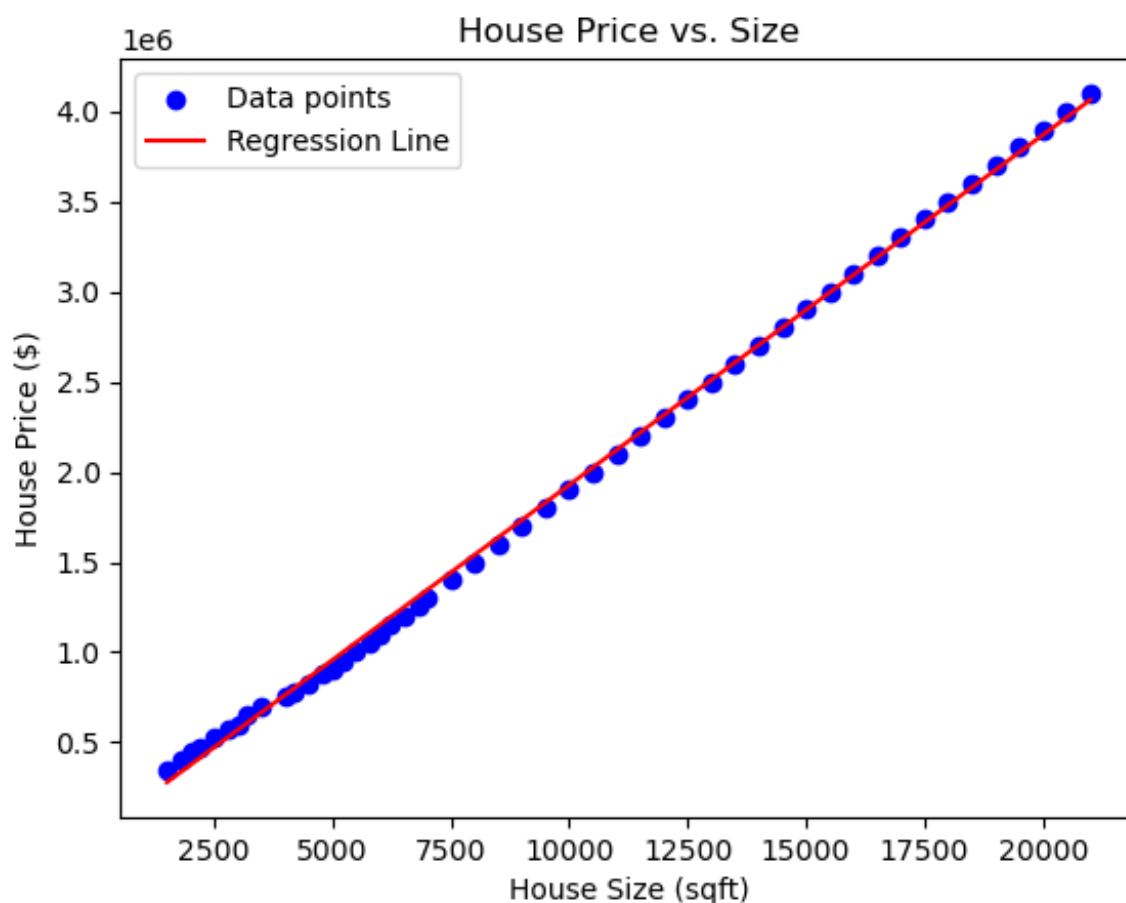
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Out[6]: array([ 957571.17449272, 3092324.31941217, 2219016.2146724 ,
3674529.72257202, 1151639.64221267, 3965632.42415194,
1830879.2792325 , 1733845.04537252, 2413084.68239235,
1248673.87607265])
```

```
In [7]: r2_score = r2_score(Y_test, Y_pred)
r2_score
print(f"R-squared Score:", r2_score)
```

R-squared Score: 0.9987913548048982

```
In [8]: plt.scatter(X,Y, color='blue', label='Data points')
plt.plot(X, model.predict(X), color='red', label = 'Regression Line')
plt.xlabel('House Size (sqft)')
plt.ylabel('House Price ($)')
plt.title('House Price vs. Size')
```

```
plt.legend()  
plt.show()
```



```
In [9]: user_House_Size = int(input("Enter House Size to predict salary: "))  
user_df = pd.DataFrame([[user_House_Size]], columns=['House Size'])  
predicted_salary = model.predict(user_df)  
print(f"The predicted House Price for House Size {user_House_Size}
```

The predicted House Price for House Size 6700 is: [1287487.56961664]